

Total No. of Questions : 6]

SEAT No. :

P4873

[Total No. of Pages : 2

B.E./Insem. - 60
B.E. (Information Technology)
INFORMATION AND CYBER SECURITY
(2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume Suitable data if necessary.*

- Q1)** a) Distinguish between Substitution and transposition ciphers. [5]
b) List and briefly define categories of security services. [5]

OR

- Q2)** a) Determine the value of x using Chinese remainder theorem. [6]
$$X = 1 \pmod{5}$$
$$X = 6 \pmod{7}$$
$$X = 8 \pmod{11}$$

b) Discuss various attacks threatening integrity. [4]

- Q3)** Explain block cipher modes of operation (ECB, CBC, CFB, OFB and counter mode) with help of block diagram. [10]

OR

- Q4)** a) Describe advantages and disadvantages of DES algorithm. [6]
b) What is the significance of extended Euclidian algorithm with reference to RSA algorithm? Illustrate. [4]

P.T.O.

Q5) a) Let the given data be - Prime numbers $p = 11$, $q = 19$ and the plain text to be sent is 40. Assume public key e as 23. Using RSA algorithm determine the cipher text for the given plain text. Also perform the reverse process of finding the plain text from the cipher text. **[6]**

b) Compare and contrast MD5 and SHA1 **[4]**

OR

Q6) a) Explain man-in-the-middle attack in Diffie-Hellman key exchange. **[6]**

b) Discuss the key management with respect to following issues. **[4]**

i) Key generation

ii) Key distribution

iii) Key updation

